

Event Handling in JavaScript

Event Delegation

Event Delegation is a technique where you attach a single event listener to a parent element to manage events on its child elements using event bubbling. This reduces memory usage and allows handling dynamically added child elements.

Example:

```
<ul id="menu">
  <li data-id="1">Home</li>
  <li data-id="2">About</li>
  <li data-id="3">Contact</li>
</ul>

<script>
const menu = document.getElementById("menu");
menu.addEventListener("click", function(event) {
  if(event.target.tagName === "LI") {
    console.log("You clicked:", event.target.innerText);
  }
});
</script>
```

Use Cases

- Handling clicks on lists, tables, or menus.
- Form validation (listen for input events on the form instead of each field).
- Dynamic UIs where elements are added/removed frequently (e.g., chat apps, todo lists).

Event Bubbling vs Event Capturing

When an event occurs, it can propagate in two ways: • Event Bubbling: Event moves from target element upwards to ancestors. • Event Capturing: Event moves from document downwards to target element.

Feature	Event Bubbling (Default)	Event Capturing
Direction	From target → ancestors (bottom-up)	From document → target (top-down)
Default in JS	Yes	No (must pass true in addEventListener)
Use Case	Event delegation, general UI handling	Intercept before target handles event (e.g., modal overlay)
Example	Click on li bubbles to ul	Overlay intercepts clicks before button

Common Interview Questions

Q: What is Event Delegation?

A: Event Delegation is attaching a single event listener to a parent to handle events for its child elements using event bubbling.

Q: Why is Event Delegation useful?

A: It saves memory, improves performance, and works with dynamically added elements.

Q: What is Event Bubbling?

A: An event starts at the target element and bubbles up to its ancestors.

Q: What is Event Capturing?

A: An event starts from the document/root and trickles down to the target element.

Q: Which phase is default in JS?

A: Event Bubbling is the default phase.

Q: How do you enable capturing?

A: Pass true as the third argument in `addEventListener`.

Q: Can you stop event propagation?

A: Yes, using `event.stopPropagation()` or `event.stopImmediatePropagation()`.

Q: Give real-world use cases of delegation.

A: Menu click handling, form validation, and managing dynamic elements efficiently.